

5.1 Definition and triangle inequality

In this section we give the basic definitions of Wasserstein distances and spaces in $\Omega \subset \mathbb{R}^d$. We will see how to adapt the analysis to the case of more general metric spaces. As we said above, we restrict our analysis to the set

$$\mathcal{P}_p(\Omega) := \left\{ \mu \in \mathcal{P}(\Omega) : \int |x|^p d\mu < +\infty \right\}.$$

Note that $p < q$ implies $\mathcal{P}_p(\Omega) \subset \mathcal{P}_q(\Omega)$ and that, whenever Ω is bounded, $\mathcal{P}_p(\Omega) = \mathcal{P}(\Omega)$.

For $\mu, \nu \in \mathcal{P}_p(\Omega)$, let us define

$$W_p(\mu, \nu) := \min \left\{ \left(\int_{\Omega \times \Omega} |x - y|^p d\gamma : \gamma \in \Pi(\mu, \nu) \right)^{\frac{1}{p}}, \right.$$

i.e. the p -th root of the minimal transport cost $\mathcal{T}_c$ for the cost $c(x, y) = |x - y|^p$. The assumption $\mu, \nu \in \mathcal{P}_p(\Omega)$ guarantees finiteness of this value since $|x - y|^p \leq C(|x|^p + |y|^p)$, whence $W_p^p(\mu, \nu) \leq C(\int |x|^p d\mu + \int |x|^p d\nu)$.

Note that, due to Jensen inequality, since all $\gamma \in \Pi(\mu, \nu)$ are probability measures, for $p \leq q$ we have

$$\left(\int |x - y|^p d\gamma \right)^{\frac{1}{p}} = \|x - y\|_{L^p(\gamma)} \leq \|x - y\|_{L^q(\gamma)} = \left(\int |x - y|^q d\gamma \right)^{1/q},$$

which implies $W_p(\mu, \nu) \leq W_q(\mu, \nu)$. In particular $W_1(\mu, \nu) \leq W_p(\mu, \nu)$ for every $p \geq 1$. We will not define here W_∞ , but we refer to Section 5.5.1 for a definition as a limit for $p \rightarrow \infty$.

On the other hand, for bounded Ω an opposite inequality also holds, since

$$\left(\int |x - y|^p d\gamma \right)^{\frac{1}{p}} \leq \text{diam}(\Omega)^{\frac{p-1}{p}} \left(\int |x - y| d\gamma \right)^{\frac{1}{p}}.$$

This implies

$$W_p(\mu, \nu) \leq C W_1(\mu, \nu)^{\frac{1}{p}}, \quad (5.1)$$

for $C = \text{diam}(\Omega)^{(p-1)/p}$.

Proposition 5.1. *The quantity W_p defined above is a distance over $\mathcal{P}_p(\Omega)$.*

Proof. First, let us note that $W_p \geq 0$. Then, we also remark that $W_p(\mu, \nu) = 0$ implies that there exists $\gamma \in \Pi(\mu, \nu)$ such that $\int |x - y|^p d\gamma = 0$. Such a $\gamma \in \Pi(\mu, \nu)$ is concentrated on $\{x = y\}$. This implies $\mu = \nu$ since, for any test function ϕ we have

$$\int \phi d\mu = \int \phi(x) d\gamma = \int \phi(y) d\gamma = \int \phi d\nu.$$

We need now to prove the triangle inequality. We give two different proofs of this fact. The first (Lemma 5.3) is easier and uses transport maps and approximation; the second (Lemma 5.4) is more general but requires a trickier tool, i.e. disintegrations of measures. □ □

Lemma 5.2. *Given $\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d)$ and χ_ε any even regularizing kernel in L^1 with $\int_{\mathbb{R}^d} \chi_\varepsilon(z) dz = 1$ and $\chi_\varepsilon(z) = \varepsilon^{-d} \chi_1(z/\varepsilon)$, then we have*

1. *the inequality $W_p(\mu * \chi_\varepsilon, \nu * \chi_\varepsilon) \leq W_p(\mu, \nu)$*
2. *the limit $\lim_{\varepsilon \rightarrow 0} W_p(\mu * \chi_\varepsilon, \nu * \chi_\varepsilon) = W_p(\mu, \nu)$.*

Proof. Take an optimal transport plan $\gamma \in \Pi(\mu, \nu)$ and define a transport plan $\gamma_\varepsilon \in \Pi(\mu * \chi_\varepsilon, \nu * \chi_\varepsilon)$ through

$$\int_{\mathbb{R}^d \times \mathbb{R}^d} \phi(x, y) d\gamma_\varepsilon := \int_{\mathbb{R}^d \times \mathbb{R}^d} \int_{\mathbb{R}^d} \phi(x+z, y+z) \chi_\varepsilon(z) dz d\gamma(x, y).$$

We need to check that its marginals are actually $\mu * \chi_\varepsilon$ and $\nu * \chi_\varepsilon$. Just consider

$$\begin{aligned} \int_{\mathbb{R}^d \times \mathbb{R}^d} \phi(x, y) d\gamma_\varepsilon &= \int_{\mathbb{R}^d \times \mathbb{R}^d} \int_{\mathbb{R}^d} \phi(x+z, y+z) \chi_\varepsilon(z) dz d\gamma(x, y) \\ &= \int_{\mathbb{R}^d \times \mathbb{R}^d} (\phi * \chi_\varepsilon)(x, y) d\gamma(x, y) = \int_{\mathbb{R}^d} (\phi * \chi_\varepsilon)(x) d\mu(x) \\ &= \int_{\mathbb{R}^d} \phi(x) d(\mu * \chi_\varepsilon)(x) \end{aligned}$$

(we use the fact that χ_ε is even to pass the convolution from ϕ to μ). The computation for the second marginal is the same. It is then easy to show that $\int |x-y|^p d\gamma_\varepsilon = \int |x-y|^p d\gamma$, since

$$\int |x-y|^p d\gamma_\varepsilon = \int_{\mathbb{R}^d \times \mathbb{R}^d} \int_{\mathbb{R}^d} |(x+z)-(y+z)|^p \chi_\varepsilon(z) dz d\gamma(x, y) = \int |x-y|^p d\gamma.$$

This shows the first part of the statement and also

$$\limsup_{\varepsilon \rightarrow 0} W_p(\mu * \chi_\varepsilon, \nu * \chi_\varepsilon)^p \leq \limsup_{\varepsilon \rightarrow 0} \int |x-y|^p d\gamma_\varepsilon = \int |x-y|^p d\gamma.$$

One can also obtain the opposite inequality by a standard semi-continuity argument. Consider $\mu_\varepsilon := \mu * \chi_\varepsilon$ and $\nu_\varepsilon := \nu * \chi_\varepsilon$, and use only the weak convergences $\mu_\varepsilon \rightharpoonup \mu$ and $\nu_\varepsilon \rightharpoonup \nu$. Call $\overline{\gamma}_\varepsilon$ the optimal plans in $\Pi(\mu_\varepsilon, \nu_\varepsilon)$. This sequence of plans is tight, since its marginals are tight (look at the proof of Theorem 1.7). First fix a sequence $\varepsilon_k \rightarrow 0$ such that $\lim_k W_p(\mu_{\varepsilon_k}, \nu_{\varepsilon_k}) = \liminf_{\varepsilon \rightarrow 0} W_p(\mu_\varepsilon, \nu_\varepsilon)$. Then extract a subsequence ε_{k_j} so as to guarantee that the optimal transport plans $\overline{\gamma}_{\varepsilon_{k_j}}$ have a weak limit $\overline{\gamma}_0$. This weak limit must belong to $\Pi(\mu, \nu)$ (the fact that the marginals of $\overline{\gamma}_0$ are μ and ν follows by the properties of composition with continuous functions of the weak convergence). Then we have

$$W_p^p(\mu, \nu) \leq \int |x-y|^p d\overline{\gamma}_0 \leq \liminf_j \int |x-y|^p d\overline{\gamma}_{\varepsilon_{k_j}} = \liminf_{\varepsilon \rightarrow 0} W_p(\mu_\varepsilon, \nu_\varepsilon),$$

where the first inequality follows from the fact that $\overline{\gamma}_0$ is not necessarily optimal but is admissible and the second follows by semi-continuity (Lemma 1.6). $\square$ $\square$

Then, we can perform a proof of the triangle inequality based on the use of optimal transport maps.

Lemma 5.3. *The quantity W_p satisfies the triangle inequality (proof by transport maps and approximation).*

Proof. First consider the case where μ and ρ are absolutely continuous and ν is arbitrary. Let T be the optimal transport from μ to ρ and S the optimal one from ρ to ν . Then $S \circ T$ is an admissible transport from μ to ν , since $(S \circ T)_\# \mu = S_\#(T_\# \mu) = S_\# \rho = \nu$. We have

$$\begin{aligned} W_p(\mu, \nu) &\leq \left(\int |S(T(x)) - x|^p d\mu(x) \right)^{\frac{1}{p}} = \|S \circ T - \text{id}\|_{L^p(\mu)} \\ &\leq \|S \circ T - T\|_{L^p(\mu)} + \|T - \text{id}\|_{L^p(\mu)}. \end{aligned}$$

Moreover,

$$\|S \circ T - T\|_{L^p(\mu)} = \left(\int |S(T(x)) - T(x)|^p d\mu(x) \right)^{\frac{1}{p}} = \left(\int |S(y) - y|^p d\rho(y) \right)^{\frac{1}{p}}$$

and this last quantity equals $W_p(\rho, \nu)$. Moreover, $\|T - \text{id}\|_{L^p(\mu)} = W_p(\mu, \rho)$, whence

$$W_p(\mu, \nu) \leq W_p(\mu, \rho) + W_p(\rho, \nu).$$

This gives the proof when $\mu, \rho \ll \mathcal{L}^d$. For the general case, first write the triangle inequality for $\mu * \chi_\varepsilon$, $\rho * \chi_\varepsilon$ and $\nu * \chi_\varepsilon$, then pass to the limit as $\varepsilon \rightarrow 0$ using Lemma 5.2. $\square$ $\square$

The proof above strongly uses the results about transport maps from Chapter 1, and is somehow specific to $\mathbb{R}^d$ (in a general metric space the tricky part would be to approximate arbitrary measures with measures such that the Wasserstein distance can be computed with maps instead of plans³). To give a more general result, we provide a different proof, which may happen to be more difficult for the reader who is not accustomed to disintegrations of measures (see Chapter 2).

Lemma 5.4. *The quantity W_p satisfies the triangle inequality (proof by disintegrations).*

Proof. Let us take μ, ρ and $\nu \in \mathcal{P}_p(\Omega)$, $\gamma^+ \in \Pi(\mu, \rho)$ and $\gamma^- \in \Pi(\rho, \nu)$. We can also choose $\gamma^\pm$ to be optimal. Let us use the Lemma 5.5 below to say that there exists a measure $\sigma \in \mathcal{P}(\Omega \times \Omega \times \Omega)$ such that $(\pi_{x,y})_\# \sigma = \gamma^+$ and $(\pi_{y,z})_\# \sigma = \gamma^-$, where $\pi_{x,y}$ and $\pi_{y,z}$ denote the projections on the two first and two last variables, respectively. Let us take $\gamma := (\pi_{x,z})_\# \sigma$. By composition of the projections, it is easy to see that $(\pi_x)_\# \gamma = (\pi_x)_\# \sigma = (\pi_x)_\# \gamma^+ = \mu$ and, analogously, $(\pi_z)_\# \gamma = \nu$. This means $\gamma \in \Pi(\mu, \nu)$

³It is not strictly necessary to use absolutely continuous measures: we actually need atomless measures, and then we can apply Theorem 1.33.

and

$$\begin{aligned}
W_p(\mu, \nu) &\leq \left(\int |x-z|^p d\gamma \right)^{\frac{1}{p}} = \left(\int |x-z|^p d\sigma \right)^{\frac{1}{p}} = \|x-z\|_{L^p(\sigma)} \\
&\leq \|x-y\|_{L^p(\sigma)} + \|y-z\|_{L^p(\sigma)} = \left(\int |x-y|^p d\sigma \right)^{\frac{1}{p}} + \left(\int |y-z|^p d\sigma \right)^{\frac{1}{p}} \\
&= \left(\int |x-y|^p d\gamma^+ \right)^{\frac{1}{p}} + \left(\int |y-z|^p d\gamma^- \right)^{\frac{1}{p}} = W_p(\mu, \rho) + W_p(\rho, \nu).
\end{aligned}$$

The proof is concluded. $\square$ $\square$

The following lemma is usually known as “gluing lemma” and allows to produce a sort of “composition” of two transport plans, as if they were maps.

Lemma 5.5. *Given two measures $\gamma^+ \in \Pi(\mu, \rho)$ and $\gamma^- \in \Pi(\rho, \nu)$ there exists (at least) a measure $\sigma \in \mathcal{P}(\Omega \times \Omega \times \Omega)$ such that $(\pi_{x,y})_{\#}\sigma = \gamma^+$ and $(\pi_{y,z})_{\#}\sigma = \gamma^-$, where $\pi_{x,y}$ and $\pi_{y,z}$ denote the projections on the two first and two last variables, respectively.*

Proof. Start by taking γ^+ and disintegrate it w.r.t. the projection π_y (see Box 2.2 in Section 2.3). We get a family of measures $\gamma_y^+ \in \mathcal{P}(\Omega)$ (as we pointed out in the same box, we can think of them as measures over Ω , instead of viewing them as measures over $\Omega \times \{y\} \subset \Omega \times \Omega$). They satisfy (and are actually defined by)

$$\int_{\Omega \times \Omega} \phi(x, y) d\gamma^+(x, y) = \int_{\Omega} d\rho(y) \int_{\Omega} \phi(x, y) d\gamma_y^+(x),$$

for every measurable function ϕ of two variables. In the same way, one has a family of measures $\gamma_y^- \in \mathcal{P}(\Omega)$ such that for every ϕ we have

$$\int_{\Omega \times \Omega} \phi(y, z) d\gamma^-(y, z) = \int_{\Omega} d\rho(y) \int_{\Omega} \phi(y, z) d\gamma_y^-(z).$$

For every y take now $\gamma_y^+ \otimes \gamma_y^-$, which is a measure over $\Omega \times \Omega$. Define σ through

$$\int_{\Omega \times \Omega \times \Omega} \phi(x, y, z) d\sigma(x, y, z) := \int_{\Omega} d\rho(y) \int_{\Omega \times \Omega} \phi(x, y, z) d(\gamma_y^+ \otimes \gamma_y^-)(x, z).$$

It is easy to check that, for ϕ depending only on x and y , we have

$$\begin{aligned}
\int_{\Omega^3} \phi(x, y) d\sigma &= \int_{\Omega} d\rho(y) \int_{\Omega \times \Omega} \phi(x, y) d(\gamma_y^+ \otimes \gamma_y^-)(x, z) \\
&= \int_{\Omega} d\rho(y) \int_{\Omega} \phi(x, y) d\gamma_y^+(x) = \int \phi d\gamma^+.
\end{aligned}$$

This proves $(\pi_{x,y})_{\#}\sigma = \gamma^+$ and the proof of $(\pi_{y,z})_{\#}\sigma = \gamma^-$ is analogous. $\square$ $\square$

Remark 5.6. All the analysis of this section can be performed on a general Polish space X instead of a subset $\Omega \subset \mathbb{R}^d$. The only differences are the definition of $\mathcal{P}_p(X)$ (we need to use $\mathcal{P}_p(X) := \{\mu \in \mathcal{P}(X) : \int d(x, x_0)^p d\mu(x) < +\infty\}$), and the fact that the first proof of the triangle inequality was specific to $\mathbb{R}^d$ (but the second works in general).

Definition 5.7. Given a Polish space X , for each $p \in [1, +\infty[$ we define its Wasserstein space of order p , $\mathbb{W}_p(X)$, as the space $\mathcal{P}_p(X)$, endowed with the distance W_p .

5.2 Topology induced by W_p

In this section we want to analyze the convergence in the space $\mathbb{W}_p$, and compare it to the notion of weak convergence. First of all, we recall what we mean by “weak convergence”: as we pointed out in Box 1.3 in Chapter 1, we use this term to denote the convergence in the duality with C_b , the space of bounded continuous functions (which is often referred to as narrow convergence), and we write $\mu_n \rightharpoonup \mu$ to say that μ_n converges in such a sense to μ .

We recall that another natural notion of convergence for measures is the one in duality with the space of functions vanishing at infinity

$$C_0(X) := \{\phi \in C(X) : \text{for all } \varepsilon > 0 \text{ there is } K \subset X \text{ compact s.t. } |\phi| < \varepsilon \text{ on } X \setminus K\}.$$

It is useful to point out some facts

- When X is compact, the spaces $C_0(X)$, $C(X)$ and $C_b(X)$ coincide.
- The condition defining $\phi \in C_0(X)$ means that ϕ tends to 0 “out of the compact subsets of X ”: this means at infinity, for $X = \mathbb{R}^d$, on $\partial\Omega$ when $X = \Omega \subset \mathbb{R}^d$ is open, and it means nothing if X is itself compact.
- For every $x_0 \in X$, the existence of functions $\phi \in C_0(X)$ with $\phi(x_0) \neq 0$ implies the existence of compact neighborhoods of x_0 . In particular, the space $C_0(X)$ is interesting only for locally compact spaces X and it only consists of the zero function whenever no point of X has compact neighborhoods (for instance, in infinite dimensional Banach spaces). Indeed, the duality result stating that $\mathcal{M}(X)$ is the dual of $C_0(X)$ is only true for locally compact spaces (see Box 1.3 in Section 1.1).

The following lemma (which is true in locally compact spaces, but we only state it for subsets of $\mathbb{R}^d$) gives a helpful criterion for weak convergence.

Lemma 5.8. *When all measures μ_n and μ are probabilities over a subset $\Omega \subset \mathbb{R}^d$, the convergence $\mu_n \rightharpoonup \mu$ coincides with the convergence in the duality with $C_0(\Omega)$.*

Proof. We only need to show that if we take $\psi \in C_b(\Omega)$, $\mu_n, \mu \in \mathcal{P}(\Omega)$ and we assume $\int \phi d\mu_n \rightarrow \int \phi d\mu$ for every $\phi \in C_0(\Omega)$, then we also have $\int \psi d\mu_n \rightarrow \int \psi d\mu$. If all the measures are probability, then we may add “for free” a constant C to ψ and, since ψ is bounded, we can choose C so that $\psi + C \geq 0$. Hence $\psi + C$ is the sup of an